# 4 Discrete-Time Fourier Series

## Objective

After completing this chapter, user will be able to

1. Understand the basics of DTFS
2. Calculate the DTFS
3. Understand DTFS properties

## Introduction

1. This chapter will focus on the operation of the Fourier series on the discrete signal.
2. We will discuss the similarities of CTFS with DTFS and differences.
3. Fourier series for discrete signals is finite series in nature, opposite to the requirement of infinite series for continuous-time periodic signals.
4. There are no convergence issues mathematically.

**Discrete-Time Fourier series:**

1. DTFS is a frequency analysis tool for periodic infinite-duration discrete-time signals, which is practical because it is discrete in frequency.
2. DTFS is calculated for periodic discrete-time signals. A signal is called to be discrete-time if independent variables are taken at a discrete interval of time.
3. A discrete-time signal $x(n)$ is periodic with period N if
   $x(n) = x(n+N)$
4. For some positive integer $N$. The smallest value of $N$ for which this equation holds is the fundamental period. The fundamental frequency is $\omega_0 = 2\pi / N$ rad/sample.
5. A periodic signal $x(n)$ can be represented by a discrete-time Fourier series made up of sinusoids (or exponentials) of fundamental frequency $\omega_0 = 2\pi / N$ and its harmonics.

**The discrete-time Fourier series (DTFS) is given by:**

$$x(n) = \sum_{k=0}^{N_0-1} C_k e^{j\omega_0 n}$$

Where $\omega_0 = \frac{2\pi}{N_0}$

$N_0 \rightarrow$ Period

$$C_k = \frac{1}{N_0} \sum_{n=0}^{N_0-1} x[n] e^{-jk\omega_0 n}$$

Where $C_k$ is Fourier series coefficient or spectral coefficient.

**Difference between continuous-time and discrete-time Fourier series:**

1. DTFS is a finite series, while CTFS is infinite Fourier series.
2. Since in the discrete domain, there are finite numbers, that is why we need not worry about convergence issues, i.e., dirichlet conditions.
3. Complex Fourier series coefficients in discrete are periodic, which were non–periodic in the case of continuous-time.

**Properties of discrete-time Fourier series:**

**Linearity:**

$x_1[n] \rightarrow c_k$

$x_2[n] \rightarrow d_k$

Then $ax_1[n] + bx_2[n] \rightarrow ac_k + bd_k$

**Time-shifting:**

$x[n] \rightarrow c_k$

Then $x[n - n_o] \rightarrow e^{-jkw_o n_o} c_k$

**Frequency-shifting:**

$x[n] \rightarrow c_k$

Then $x[n] e^{M\left(\frac{2\pi}{N}\right)n} \rightarrow c_{k-M}$

**Time-reversal:**

$x[n] \longrightarrow C_k$

Then $X[-n] \longrightarrow C_{-k}$

**Time scaling property:**

$x[n] \rightarrow c_k$

Then $x\left[\frac{n}{\alpha}\right] \rightarrow c_{\alpha k}$ [$\omega_0$ remains same]

**Multiplication property:**

In time domain:

$x_1[n] \rightarrow C_k$

$x_2[n] \rightarrow d_k$

Then $x_1[n]x_2[n] \rightarrow C_k * d_k$

In frequency domain:

$X_1[n] \rightarrow C_k$ and $X_2[n] \rightarrow d_k$

Then $X_2[n] * X_2[n] \rightarrow N_0 C_k d_k$

**Conjugate property:**

$x[n] \longrightarrow C_k$

$x^*[n] \longrightarrow C^*_{-k}$

If $x[n]$ is real then $C^*_{-k} = C_k$ or $C_{-k} = C^*_k$

**Parseval's relationship:**

$x[n] \longrightarrow C_k$

Then $\sum_{n=0}^{N_0-1}[x[n]]^2 = N_0\left[\sum_{k=0}^{N_0-1}|C_k|^2\right]$

**1st difference property:**

$x[n] \rightarrow C_k$

Then $x[n] - x[n-1] \rightarrow C_k\left[1 - e^{-jk\omega_0}\right]$

**Periodicity:**

$x[n] \rightarrow C_k$

$C_{k+N_0} = C_k$

Where $N_0$ is the fundamental period of $x[n]$

It means Fourier series representation of a discrete time signal is also periodic.

**Note:** $x[n] \rightarrow C_k$

1. If $x[n]$ is real and even signal, then $C_k$ is real and even.
2. If $x[n]$ is real and odd, then $C_k$ is purely imaginary and odd.

## Magnitude and Phase Spectrum of Discrete-Time Periodic Signals (Fourier spectra)

1. The Fourier series consists of N components

   $X_0, X_1 e^{j\omega_0 n}, ....., X_2 e^{j2\omega_0 n}, ....., X_{N-1} e^{j(N-1)\omega_0 n}$
2. The frequencies of these components are $0, \omega_0, 2\omega_0 ....., (N-1)\omega_0$ where $\omega_0 = 2\pi / N$. T he amount of the K[th] harmonic is $X_k$ (the Fourier coefficient).
3. We can plot this amount $X_k$ as a function of index $k$ or $\omega$. Such a plot is called the Fourier spectrum of $x(n)$.
4. In general, the Fourier coefficients $X_k$ are complex, and they can be represented in the polar form as

   $X_k = |X_k| e^{j\angle x_k}$
5. The plot of $|X_k|$ versus $K$ is called the magnitude spectrum, and that of $\angle X_k$ versus $\omega$ is called the angle (or phase) spectrum. These two plots together are the frequency spectra of $x(n)$. Knowing these spectra, we can reconstruct or synthesise $x(n)$.

## Solved Examples

**1.** Determine the Fourier series coefficients of the signal $x(n)$ and plot its magnitude and phase spectrum

$$x(n) = 1 + \sin\left(\frac{2\pi}{N}n\right) + 3\cos\left(\frac{2\pi}{N}n\right) + \cos\left(\frac{4\pi}{N}n + \frac{\pi}{2}\right)$$

**Solution:**

The given signal is periodic with period $N$ and Frequency $\omega_0 = 2\pi / N$, and it can be written as

$$x(n) = 1 + \sin(\omega_0 n) + 3\cos(\omega_0 n) + \cos\left(2\omega_0 n + \frac{\pi}{2}\right)$$

$$x(n) = 1 + \left(\frac{e^{j\omega_0 n} - e^{-j\omega_0 n}}{2j}\right) + 3\left(\frac{e^{j\omega_0 n} + e^{-j\omega_0 n}}{2}\right) + \left(\frac{e^{j2\omega_0 n + \pi/2} + e^{-j2\omega_0 n + \pi 2}}{2}\right)$$

$$= 1 + \left(\frac{3}{2} + \frac{1}{2j}\right)e^{j\omega_0 n} + \left(\frac{3}{2} - \frac{1}{2j}\right)e^{-j\omega_0 n} + \left(\frac{1}{2}e^{j\pi/2}\right)e^{j2\omega_0 n} + \left(\frac{1}{2}e^{-j\pi/2}\right)e^{-j2\omega_0 n}$$

Comparing the above equation with

$$x(n) = X_0 + X_1 e^{j\omega_0 n} + X_{-1} e^{-j\omega_0 n} + X_2 e^{j2\omega_0 n} + X_{-2} e^{-j2\omega_0 n}$$

We get

$$X_0 = 1$$

$$X_1 = \frac{3}{2} + \frac{1}{2j} = \frac{3}{2} - \frac{1}{2}j$$

$$X_{-1} = \frac{3}{2} - \frac{1}{2j} = \frac{3}{2} + \frac{1}{2}j$$

$$X_2 = \frac{1}{2}j$$

$$X_{-2} = -\frac{1}{2}j$$

All other Fourier series coefficients in the interval of summation in the synthesis equation are zero. The magnitude of the Fourier coefficients are

$$|X_0| = 1,\ |X_1| = |X_{-1}| = \frac{\sqrt{10}}{2},$$

$$|X_2| = |X_{-2}| = \frac{1}{2}$$

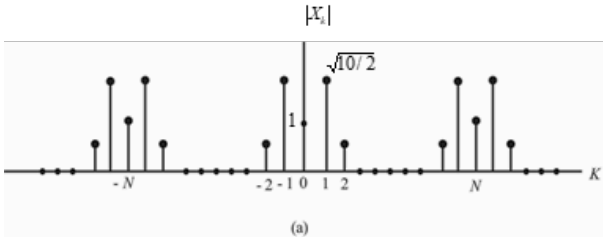
The phase of the Fourier coefficients are:

$$\angle X_0 = 0,\ \angle X_1 = -\tan^{-1}(1/3),$$

$$\angle X_{-1} = \tan^{-1}(1/3)$$

$$\angle X_2 = \frac{\pi}{2},\ \angle X_{-2} = -\frac{\pi}{2}$$

The magnitude spectrum and the phase spectrum are depicted in figs a) and b), respectively.




(b)

**Fig. 4.1 a) Magnitude spectrum, b) Phase spectrum**

**2.** Calculate Fourier series coefficient $C_k$ of the given signal $x[n]$.

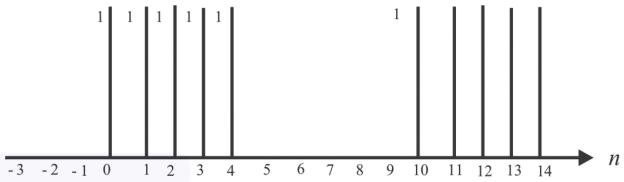



**Fig. 4.2**

**Solution:**

Given signal is periodic and the period is $N_0 = 10$

$$C_k = \frac{1}{N_0}\sum_{n=0}^{N_0-1} x[n]e^{-jk\omega n}$$

$$C_0 = \frac{1}{10}\sum_{n=0}^{9} x[n] = \frac{1}{10}[1+1+1+1+1] = 0.5$$

$$C_k = \frac{1}{10}\sum_{n=0}^{9} x[n]e^{-jk\left(\frac{2\pi}{10}\right)n} = \frac{1}{10}\sum_{n=0}^{9} x[n]e^{-jk\left(\frac{\pi}{5}\right)n}$$

$$C_k = \frac{1}{10}\left[1 + e^{-jk\left(\frac{\pi}{5}\right)} + e^{-jk\left(\frac{\pi}{5}\right)2} + e^{-jk\left(\frac{\pi}{5}\right)3} + e^{-jk\left(\frac{\pi}{5}\right)4}\right]$$

$$C_k = \frac{1}{10}\left[1 + e^{-jk\left(\frac{\pi}{5}\right)} + e^{-jk\left(\frac{\pi}{5}\right)2} + e^{-jk\left(\pi - \frac{2\pi}{5}\right)} + e^{-jk\left(\pi - \frac{\pi}{5}\right)}\right]$$

$$C_k = \frac{1}{10}\left[1 + e^{-jk\left(\frac{\pi}{5}\right)} + e^{-jk\left(\frac{2\pi}{5}\right)} - e^{jk\left(\frac{2\pi}{5}\right)} - e^{jk\left(\frac{\pi}{5}\right)}\right]$$

$$= \frac{1}{10}\left[1 - \left(e^{jk\left(\frac{\pi}{5}\right)} - e^{-jk\left(\frac{\pi}{5}\right)}\right) - \left(e^{jk\left(\frac{2\pi}{5}\right)} - e^{-jk\left(\frac{2\pi}{5}\right)}\right)\right]$$

$$C_k = \frac{1}{10}\left[1 - 2j\sin\left(\frac{k\pi}{5}\right) - 2j\sin\left(\frac{k2\pi}{5}\right)\right]$$

## Level 1 Exercise

**1.** If the Fourier series coefficient of a signal is periodic, the signal must be:
(A) Continuous-time, periodic
(B) Discrete-time, periodic
(C) Continuous-time, non-periodic
(D) Discrete-time, non-periodic

**2.** The DTFS coefficients of a real and even periodic signal are:
(A) Real and odd
(B) Imaginary and even
(C) Real and even
(D) Imaginary and odd

**3.** The DTFS coefficients of a real and odd periodic signal are:
(A) Real and odd
(B) Imaginary and even
(C) Real and even
(D) Imaginary and odd

**4.** Let $X_k$ represent the DTFS coefficients of the periodic sequence $x(n)$ with period N. The DTFS coefficients of the signal $(-1)^n x(n)$ in terms of $X_k$ are:
(A) $X_k$
(B) $X_{-k}$
(C) $X_{k+\frac{N}{2}}$
(D) $X_{k-\frac{N}{2}}$

**5.** Let $X_k$ represent the DTFS coefficients of the periodic sequence $x(n)$ with period N. The DTFS coefficients of the signal $x(n-1)$ in terms of $X_k$ are:
(A) $X_k e^{-jk2\pi/N}$
(B) $X_k e^{jk2\pi/N}$
(C) $X_{k+1}$
(D) $X_{k-1}$

## Level 1 Solutions

**1. (B)**

| $x(t)$ | FSC |
|---|---|
| Continuous-time aperiodic | ⟶ |
| Discrete time periodic | ⟶ |

**Table 4.1**

**Note:** Vice Versa is also true. Hence, if FSC is periodic, then the signal must be discrete-time, periodic.

**2. (C)**

| $x(t)$ | **DTFS** |
|---|---|
| 1. Real + Even | Real + even |
| 2. Real + odd | Imaginary + odd |
| 3. Imaginary + even | Imaginary + even |
| 4. Imaginary + odd | Real + odd |

**Table 4.2**

According to (1) option, 'C' is correct.

## Level 2 Exercise

**1. (0.5)**

$$x(n) = 1 + \omega s\left(\frac{\pi n}{8}\right)$$

$$= 1 + e^{\frac{j\pi n}{8}} + e^{\frac{-j\pi n}{8}} \quad \text{...(i)}$$

$$a_k = \frac{1}{16}\sum_{n=0}^{15} x(n) e^{\frac{-j2\pi}{16}Kn}$$

N=16

$$\therefore x(n) = \sum_{K=0}^{15} a_k e^{\frac{j2\pi}{16}Kn}$$

$$= a_0 + a_1 e^{\frac{j2\pi}{16}n} + a_{-1} e^{-j\frac{2\pi}{16}n}$$

$$= a_0 + a_1 e^{\frac{j\pi n}{8}} + a_{-1} e^{\frac{-j2\pi n}{8}} \quad ...(2)$$

Comparing (i) and (ii),

$$\therefore a_0 = 1, \qquad a_1 = a_{-1} = \frac{1}{2}$$

$\therefore$ Since, $x(n)$ is periodic with $N = 16$

$$\therefore \; X_k = \phi X_{(K+N)} = X_{(K+2N)}$$

$$\therefore X_K = X_{(K+16)} = X_{(K+32)}$$

$\therefore$ if $K = -1$,

$$\therefore X_{-1} = X_{(-1+16)} = X_{(-1+32)}$$

$$\therefore X_{-1} = X_{15} = X_{31} = \frac{1}{2}$$

**2. (C)**

The DTFS coefficients have period $N = 9, \text{so}\, \omega_0 = 2\pi / 9$. The DTFS coefficients over the interval k= -4 to k = 4 are

$$X_k = \left|X_k\right| e^{j\angle X_k}, \; -4 \le k \le 4$$

$$X_{-4} = 0, X_{-3} = 1e^{j2\pi/3}, \; X_{-2} = 2e^{j\pi/3}, \; X_{-1} = 0,$$

$$X_0 = 1e^{j\pi}, X_1 = 0, X_2 = 2e^{-j\pi/3},$$
$$X_3 = 1e^{-j2\pi/3}, \; X_4 = 0$$

It is convenient to evaluate over the interval k = -4 to k = 4 to obtain $x(n)$

$$x(n) = \sum_{k=(N)} X_k e^{jk\omega_0 n}$$

$$= \sum_{k=-4}^{4} X_k e^{jk2\pi/9n}$$

$$= X_{-4}e^{-j8\pi/9n} + X_{-3}e^{-j6\pi/9n} + X_{-2}e^{-j4\pi/9n}$$
$$+X_{-1}e^{-j2\pi/9n} + X_0 + X_1 e^{j2\pi/9n}$$
$$+X_2 e^{j4\pi/9n} + X_3 e^{j6\pi/9n} + X_4 e^{j8\pi/9n}$$

$$= e^{j2\pi/3}e^{-j6\pi/9n} + 2e^{jx/3}e^{-j4\pi/9n} + e^{jk}$$
$$+2e^{-j\pi/3}e^{j4\pi/9n}e^{-j2\pi/3}e^{j6\pi/9n}$$

$$= 2\left(\frac{e^{j(6\pi/9n-2\pi/3)} + e^{-j(6\pi/9n-2\pi/3)}}{2}\right)$$

$$+4\left(\frac{e^{j(4\pi/9n-\pi/3)} + e^{-j(4\pi/9n-\pi/3)}}{2}\right) - 1$$

$$x(n) = 2\cos\left(\frac{6\pi}{9}n - \frac{2\pi}{3}\right)$$
$$+ 4\cos\left(\frac{4\pi}{9}n - \frac{\pi}{3}\right) - 1$$

**3. (C)**

The signal has a period $N = 5, \text{so}\, \omega_0 = \frac{2\pi}{5}$, .

Also, the signal has odd symmetry, so we sum over

n= –2 to n = 2 in to obtain $X_k$.

$$X_k = \frac{1}{N}\sum_{n=(N)} x(n)e^{-jk\omega_0 n}$$

$$= \frac{1}{5}\sum_{n=-2}^{2} x(n)e^{-jk2\pi/5n}$$

$$= \frac{1}{5}\left[\begin{matrix} x(-2)e^{jk4\pi/5} + x(-1)e^{jk2\pi/5} + x(0) \\ +x(1)e^{-jk2\pi/5} + x(2)e^{-jk4\pi/5} \end{matrix}\right]$$

Using the values of x(n), we get

$$X_k = \frac{1}{5}\left[1 + \frac{1}{2}e^{jk2\pi/5} - \frac{1}{2}e^{-jk2\pi/5}\right]$$
$$= \frac{1}{5}\left[1 + j\sin\left(k\frac{2\pi}{5}\right)\right]$$

The one period of the DTFS coefficients $X_k, k = -2$ to $k = 2$ are

$$X_{-2} = \frac{1}{5} - j\frac{\sin(4\pi/5)}{5} = 0.232e^{-j0.531}$$

$$X_{-1} = \frac{1}{5} - j\frac{\sin(2\pi/5)}{5} = 0.276e^{-j0.760}$$

$$X_0 = \frac{1}{5} = 0.2e^{j0}$$

$$X_1 = \frac{1}{5} + j\frac{\sin(2\pi/5)}{5} = 0.276e^{j0.760}$$

$$X_2 = \frac{1}{5} + j\frac{\sin(4\pi/5)}{5} = 0.232e^{j0.531}$$